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A Proposal to Publish a Technical Specification for Contracts

We propose to publish a Technical Specification for Contracts based on these sources of wording:

- [P0542R5] and [P1323R2], which were incorporated into the C++20 Working Draft before being removed by [P1823R0], plus minor changes introduced by the project editor;
- [P1607R1], which was approved by EWG, and was subsequently in the process of being reviewed by CWG when Contracts were pulled from C++20; and
- [P1344R1], which was approved by CWG, but not yet applied at the time when Contracts were pulled from C++20.

Publishing a Contracts TS would pave the way for implementations to begin providing users with the ability to write contracts, which would in turn allow users to start gaining much wider real-world experience. We hope that such experience will more expeditiously result in consensus over what the Contracts feature should comprise so as to be most useful to a wider range of C++ programmers, which to date has been manifestly difficult to achieve.

We note that, as of now, SG21 has not produced a complete proposal, and is on track to produce only a minuscule MVP that itself has been stalled for going on a year now. In contrast, most of the features in the proposed Contracts TS already have considerable implementation experience (see [P1680R0]). We believe that publishing a TS would be of enormous value in helping to advance the state of Contracts so that we can land a useful viable product in C++26.

The initial draft of the proposed Contracts TS will follow in a separate paper, [P2660R0]. We have deliberately kept it as close as possible to the aforementioned sources, making changes only where strictly necessary to produce a coherent whole that is based on the C++23 DIS, in order to have a starting point that consists only of features that were already approved by EWG¹. Another, separate paper, [P2661R0], will propose to incorporate evolution since P1607R1.

¹One notable exception is that we have chosen to rename the `inform` contract behavior to `observe` so that now, each of the four contract behavior names is a verb that has the contract as its direct object.

References

- [P0542R5] G. Dos Reis, J. D. Garcia, J. Lakos, A. Meredith, N. Myers, and B. Stroustrup, *Support for contract based programming in C++*
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- [P1680R0] Andrew Sutton and Jeff Chapman, *Implementing Contracts in GCC*
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- [P1823R0] Nicolai Josuttis, Ville Voutilainen, Roger Orr, Daveed Vandevoorde, John Spicer, and Christopher Di Bella, *Remove Contracts from C++20*
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